

Python Data Structures & Collections Guide

1. Advanced Collections

The 'collections' module provides specialized container datatypes:

- defaultdict: handles missing keys seamlessly
- Counter: counts element occurrences
- deque: double-ended queue for fast appends/pops
- namedtuple: tuples with named fields

2. List & Dict Comprehensions

Comprehensions allow inline creation of lists and dicts:

- `[x**2 for x in range(10) if x % 2 == 0]`
- `{k: v for k, v in zip(keys, values)}`